Alexandra McDaniel

(775) 636-1009 | mcdaniel.alexh@gmail.com | www.linkedin.com/in/alexh-mcdaniel
https://github.com/alexhoppsalot/mcdaniel-portfolio.git

PERSONAL PROFILE

Graduate student in Electrical Engineering (GPA 4.0) with research experience in power systems and underwater acoustics. Skilled in Python, MATLAB, and project management. Seeking opportunities to apply technical and analytical expertise to energy systems and reliability engineering.

EDUCATION

Master's in Electrical Engineering | Laramie, WY — Expected Graduation: May 2026
University of Wyoming
GPA 4.0

Bachelor's in Applied Physics | Provo, UT — July 2024
Brigham Young University
GPA 3.75

Rancho High School | Las Vegas, NV — June 2018
Clark County School District
GPA 4.8 (weighted) | Valedictorian

WORK EXPERIENCE

Graduate Research Assistant | Laramie, WY — August 2024 – Present
University of Wyoming Dept. of Electrical Engineering
- Analyzed power quality disturbances and their impact on renewable-integrated transmission systems, supporting ongoing research in grid reliability

Graduate Teaching Assistant | Laramie, WY — August 2024 – Present
University of Wyoming Dept. of Electrical Engineering
- Taught a section of Basic Circuits (EE 2210) and Communication Theory (EE 4440) for engineering students

Independent Research Contractor | Remote — August 2023 – Present
Brigham Young University Hydroacoustic Lab
- Trained deep learning models for seabed classification from merchant-ship acoustic sources
- Analyzed the seabed classification precision of a robust deep learning model that accounts for variation in ocean sound speed

Knobles Scientific and Analysis
- Discovered deep sediment heterogeneity in the New England Mudpatch by inferring seabed composition from merchant-ship acoustic sources
- Published a paper in JASA for this work

** Funded by the Office of Naval Research*

Substitute Teacher | Laramie, WY — August 2023 – May 2024
Albany County School District
- Taught at schools K-12 in Laramie, WY

Field Technician | Laramie, WY — May – August 2024
University of Wyoming Dept. of Botany
- Assisted with field surveys of vegetation and water potential in plants while leading the in-lab preparation and calibration of electrical field equipment
- Successfully set up two free-standing meteorological towers in the field and led troubleshooting tasks

Undergraduate Research Assistant | Provo, UT January 2022 – July 2023
Brigham Young University Dept. of Physics and Astronomy
- Characterized a laboratory water tank for sound measurements in a model ocean environment
- Introduced temperature and sound speed variability to better model the sound speed environments found in the shallow ocean

Physics Lab Manager | Provo, UT January 2022 – July 2023
Brigham Young University Dept. of Physics and Astronomy
- Managed a team of six undergraduates and coordinated lab setup for eight undergraduate physic courses, ensuring smooth delivery of experiments for 2,000+ students each semester
- Implemented the SCRUM and AGILE methodologies for team management

Physics Lab Technician | Provo, UT January – September 2019, May– December 2021
Brigham Young University Dept. of Physics and Astronomy
- Setup, test, and trouble shoot equipment for eight undergraduate physics courses
- Led the re-design of two introductory labs for undergraduate students
- Maintaining and making small repairs to physics equipment used for lab classes

Lab Teaching Assistant | Provo, UT September 2021 – December 2021
Brigham Young University Dept. of Physics and Astronomy
- Taught a two-hour lab once a week
- Helped students trouble shoot equipment and understand correct experimental practices
- Graded lab reports

FIELD EXPERIENCE

Seabed Characterization Experiment 2022 Falmouth, MA
Student Scientist May 2022
- Assisted with conductivity-temperature-depth measurements while aboard the RV Neil Armstrong

SKILLS

- Programming, Software, & OS:
    - Proficient: Python, MATLAB, Github, Visual Studio Code, Overleaf LaTeX Editor
    - Intermediate: MATLAB Simulink, LabVIEW, Mathematica
    - Beginner: C++, CRBasic, R, Linux
- Engineering tools:
    - Intermediate: RT-Lab
    - Beginner: SolidWorks, Autodesk Inventor
- Technical skills:
    - Experience: soldering, machining, photolithography, vacuum design
- Project and Team Management:
    - Methods: SCRUM, AGILE, Trello, ClickUp
    - Experience: scheduling, hiring, interviewing, professional presentations
- Bilingual: English / Spanish

PUBLICATIONS

- Hopps-McDaniel, Alexandra M., et al. "Deep sediment heterogeneity inferred using very low-frequency features from merchant ships." The Journal of the Acoustical Society of America 156.4 (2024): 2265-2274.
- Hopps-McDaniel, Alexandra M., and Tracianne B. Neilsen. "Temperature-induced sound speed variability in a laboratory water tank." Proceedings of Meetings on Acoustics. Vol. 51. No. 1. AIP Publishing, 2023.
- Harmer, Madeline, et al. "Leveraging scientific modeling to engage pre-med undergraduates in physics lab courses." Physical Review Physics Education Research 20.2 (2024): 020150.
- Undergraduate Capstone Project: Hopps, Alexandra, “Characterizing the sound speed variability in laboratory water tank,” Advisor: Traci Neilsen (Capstone, Jun 2023).

PRESENTATIONS AT CONFERENCES

- May 2025 - Acoustical Society of America (New Orleans, Louisiana) | "The impact of water sound speed on deep learning seabed classification from shipping noise"
- May 2024 - Acoustical Society of America (Ottawa, Canada) | "Deep sediment characterization from very low-frequency features from merchant ships"
- May 2023 - Acoustical Society of America (Chicago, Illinois) | "Introducing sound speed variability in a laboratory water tank"
- February 2023 - Brigham Young University Student Research Conference (Provo, Utah) | "Introducing sound speed variability in a laboratory water tank"
- July 2022 - American Association for Physics Teachers (Grand Rapids, Michigan) | "Physics labs that resonate with pre-medicine students"

POSTER SESSIONS

- March 2024 – Seabed Characterization Experiment Workshop (Kingston, Rhode Island) | "Deep sediment characterization from very low-frequency features from merchant ships"
- January 2023 - Conference for Undergraduate Women in Physics (Santa Cruz, California) | "Introducing sound speed variability in a laboratory water tank"

AWARDS

- 2024: University of Wyoming full graduate assistantship in electrical engineering
- 2023: University of Wyoming full graduate assistantship in electrical engineering
- 2022: McKay Family Foundation - BYU Women in STEM Full Tuition Scholarship
- 2018: Brigham Young University Full Tuition Scholarship
- 2018: Valedictorian (class of 850 students)
- 2018: Career and Technical Education Award in Aerospace Engineering
- 2018: Second place in Chemistry Lab at the Nevada State Science Olympiad Competition
- 2017: First place in Public Health at the Nevada State Health Occupation Students of America competition
- 2016: Black Belt in Kenpo Karate
    - Placed in the Annual Las Vegas Tournament in 2008, 2009, 2014, and 2016

RELEVANT COURSE WORK

- <u>Engineering</u>: Cyber Physical Systems, Power Reliability, Power Quality, Power Engineering, Statistical Signal Processing, Communication Theory, Thermodynamics, Biomedical Engineering
- <u>Mathematics</u>: Linear Algebra, Calculus 3, Ordinary Differential Equations, Partial Differential Equations, Theory of Statistics 1 & 2
- <u>Programming</u>: Convolutional Neural Networks, Python lab, MATLAB lab, C++
- <u>Physics</u>: Electricity and Magnetism 1 & 2, Acoustics, Mechanics, Optics, Modern Physics, Intro to Physics
- <u>TA courses</u>: Basic Circuits Lab, Communication Theory Lab, Introductory Physics Lab

UNDERGRADUATE RESEARCH PROJECTS

Three semesters of student-designing experimentation in advanced lab classes

- Built a laser microphone using principles of optics, interferometry, and a transimpedance photodiode to record and analyze songs, chirps, and white noise
- Designed a vacuum system for creation of plasma and plotting the Paschen curve of various gases
- Conducted an experiment to determine the critical temperature of a superconductor sample BiPbSrCaCuO
- Modeled how linear density affected resonance patterns in a Chladni plate

VOLUNTEER EXPERIENCE

Victim Assistance Mentor | Provo, UT — June 2021 – December 2022

Utah Children's Justice Center

- Completed a 40-hour training on sexual violence and child abuse
- Spent 2-3 hours every week doing a fun activity with my mentee

Religious Missionary | Salem, OR September 2019 – March 2021

Church of Jesus Christ of Latter-day Saints

- Served 9 hours a day for 6 days a week. Taught people from the scriptures and did regular service projects
- Learned to speak Spanish fluently and taught English as a second language

Big Sister | Las Vegas, NV September 2015 – May 2018

Big Brother Big Sister - Walter Bracken Elementary School

- Mentored an elementary school student for one hour once a week. Activities included helping with homework and playing games